

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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Assistant Professor

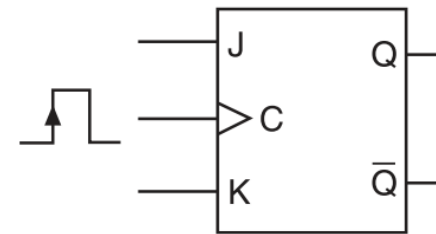
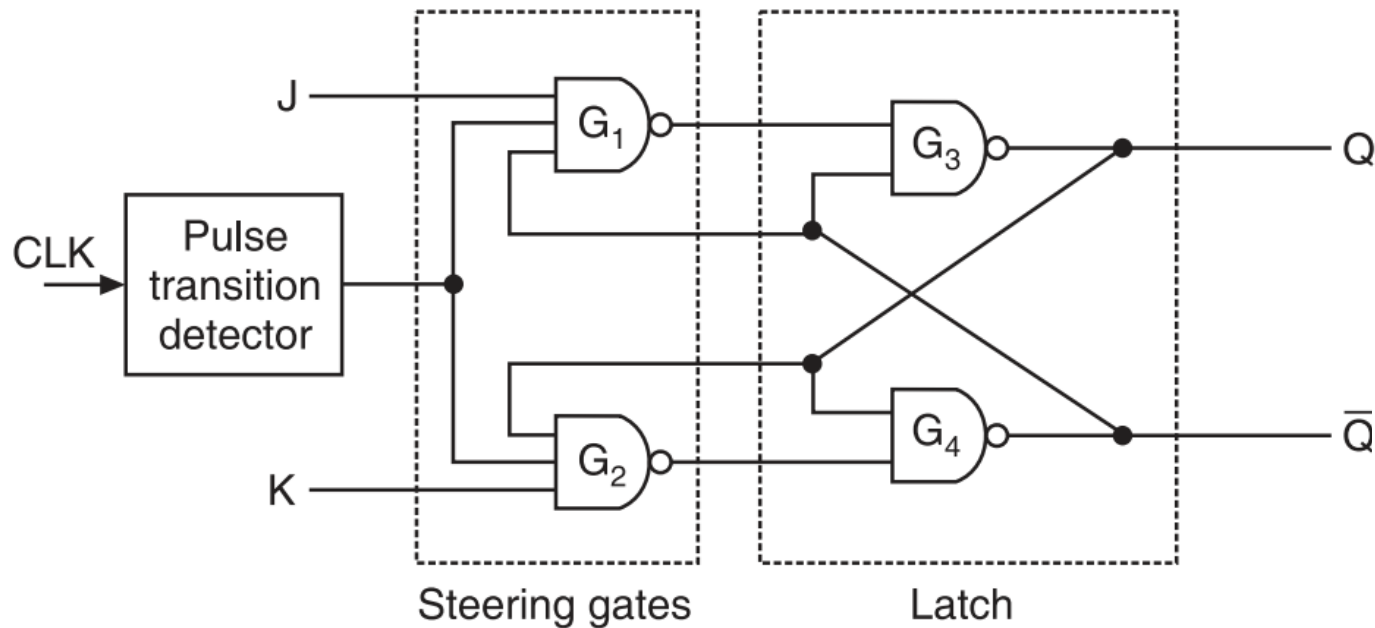
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JK Flip Flop



A J-K flip-flop is nothing more than an S-R flip-flop with an added layer of feedback. The full form of JK flip flop is Jack Kilby flip flop.



(a) Logic symbol

C	J	K	Q_n	Q_{n+1}
$\uparrow$	0	0	0	0
$\uparrow$	0	0	1	1
$\uparrow$	0	1	0	0
$\uparrow$	0	1	1	0
$\uparrow$	1	0	0	1
$\uparrow$	1	0	1	1
$\uparrow$	1	1	0	1
$\uparrow$	1	1	1	0
0	x	x	0	0
0	x	x	1	1

(b) Truth table

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Case 1: Clk = 1; J = 0; K = 0;

Output of G_1 is 1

Output of G_2 is 1

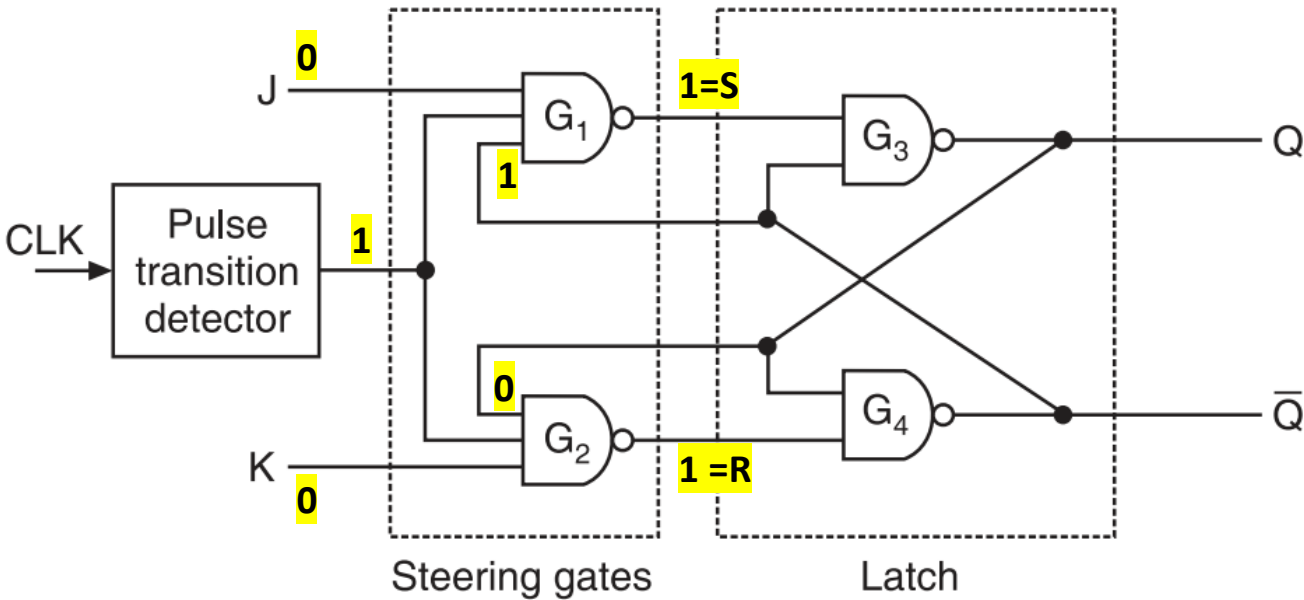
Output of G_3 is Q

Output of G_4 is $\bar{Q}$

ISM) DHANBAD

Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)



C	J	K	Q_n	Q_{n+1}
$\uparrow$	0	0	0	0
$\uparrow$	0	0	1	1

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Case 2: Clk = 1; J = 0; K = 1;

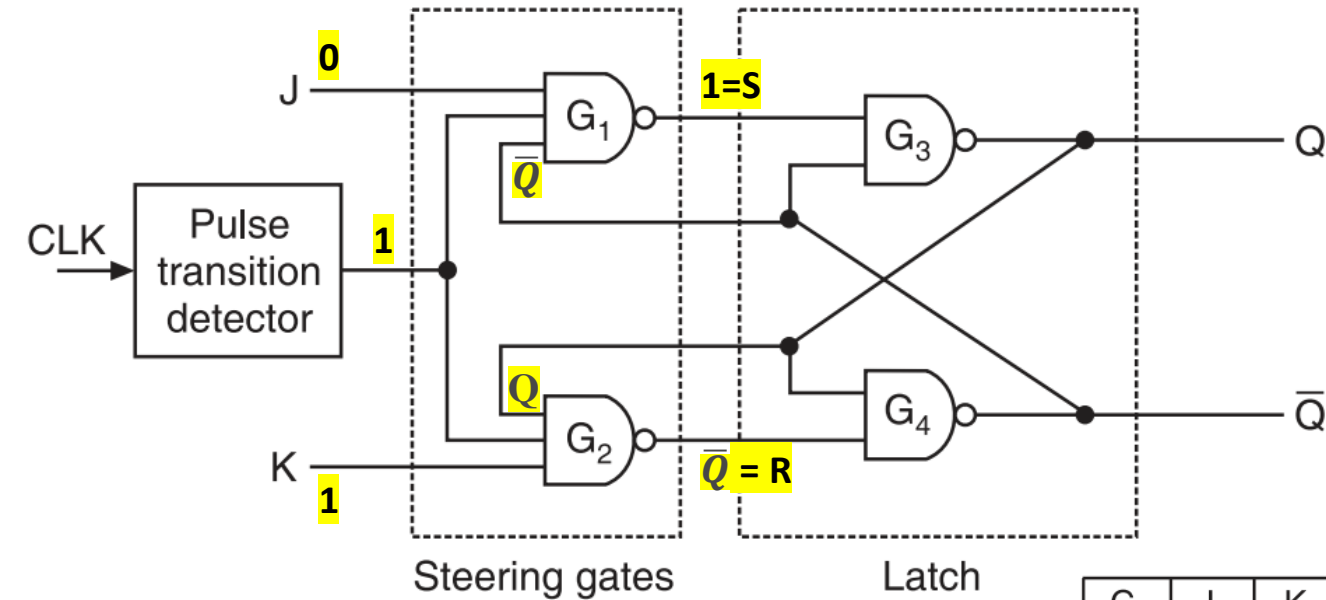
Output of G_1 is 1

Output of G_2 is $\bar{Q}$

Let us assume $Q = 1$

Output of G_3 is 0

Output of G_4 is 1



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
$\uparrow$	0	0	0	0
$\uparrow$	0	0	1	1
$\uparrow$	0	1	0	0

JK Flip Flop



Case 3: Clk = 1; J = 0; K = 1;

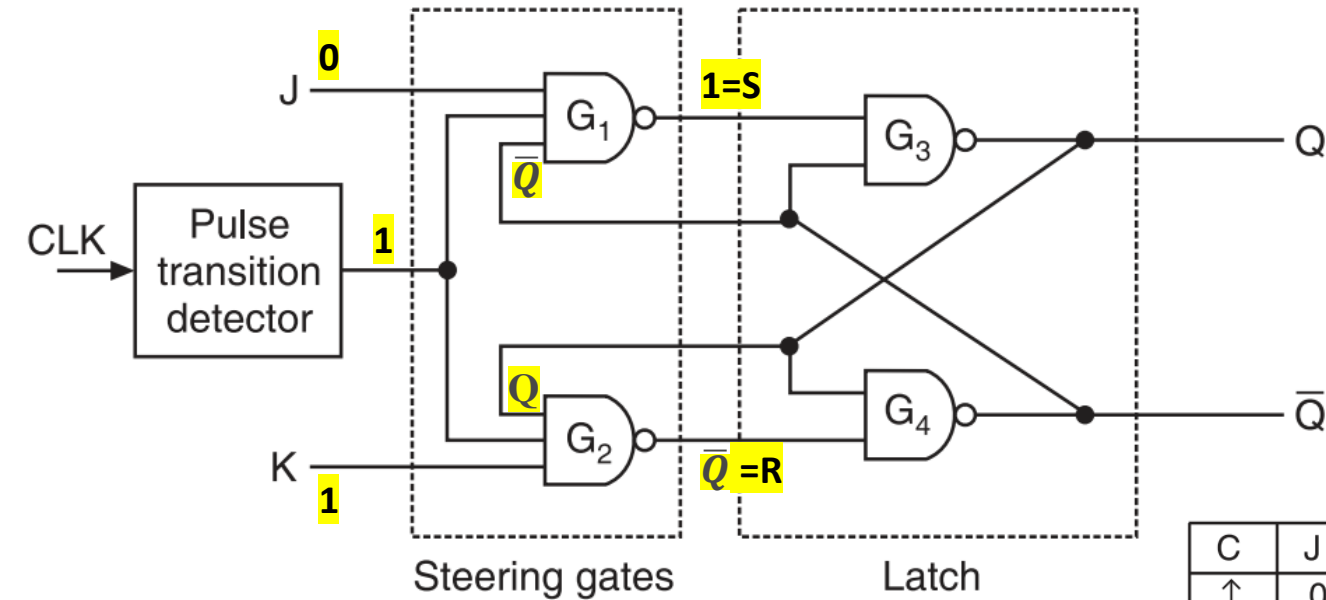
Output of G_1 is 1

Output of G_2 is $\bar{Q}$

Let us assume $Q = 0$

Output of G_3 is 0

Output of G_4 is 1



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0

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Case 4: Clk = 1; J = 1; K = 0;

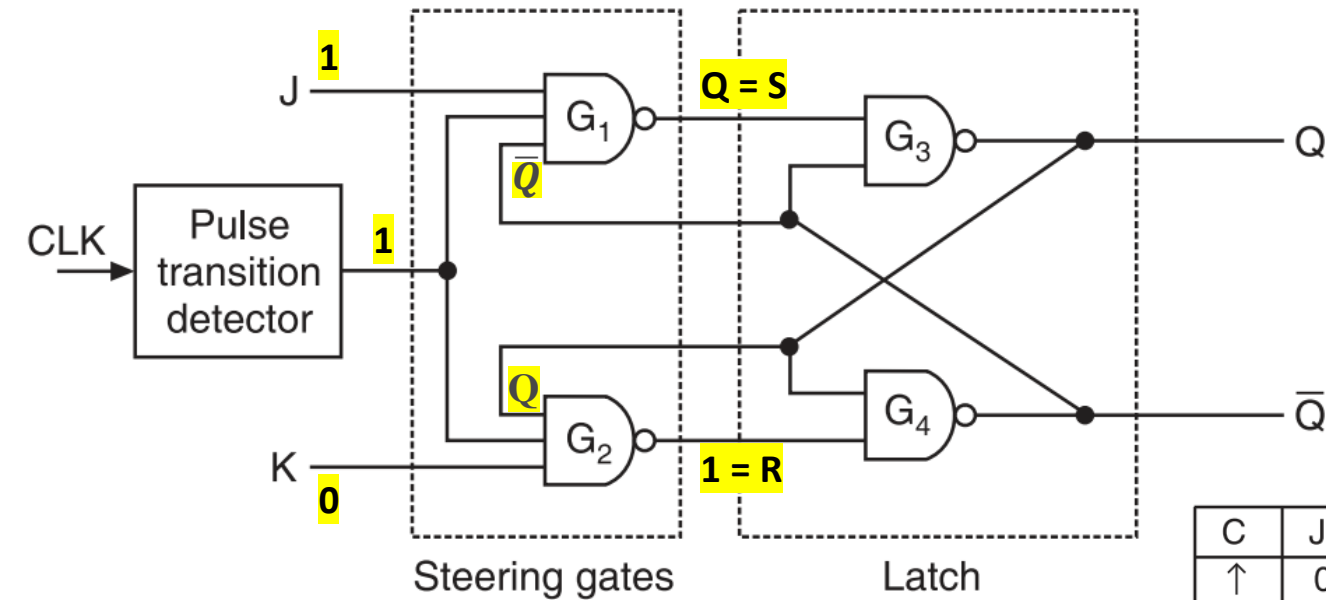
Output of G_1 is Q

Output of G_2 is 1

Let us assume $Q = 0$

Output of G_3 is 1

Output of G_4 is 0



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1

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Case 5: Clk = 1; J = 1; K = 0;

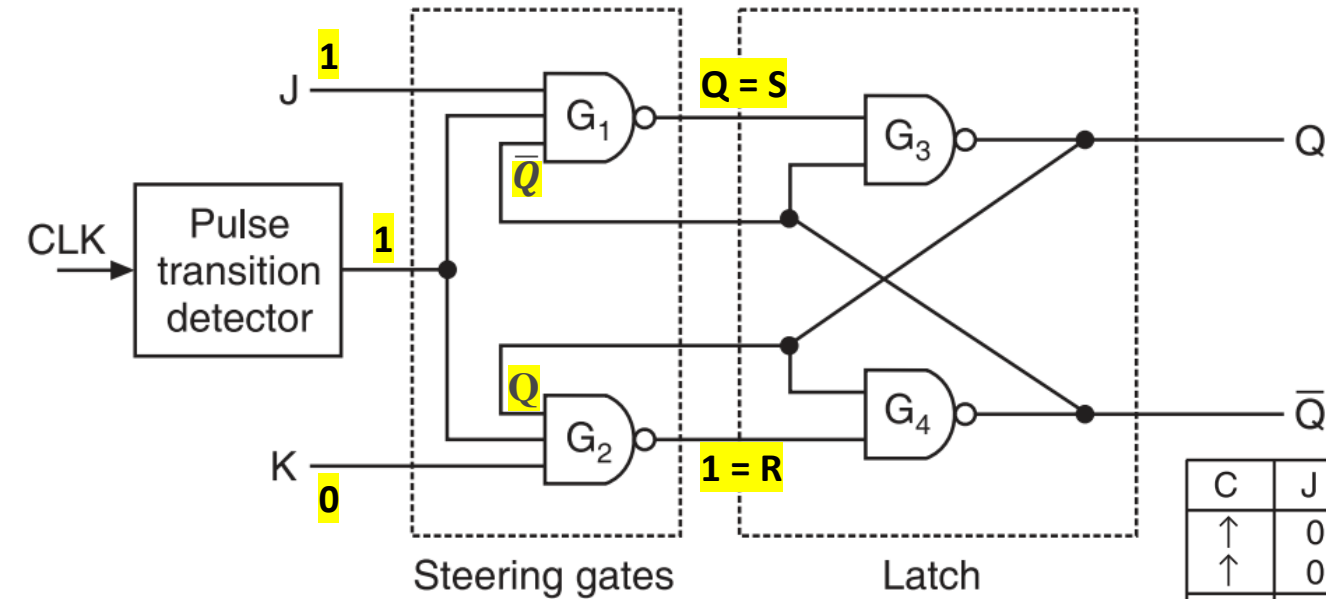
Output of G_1 is Q

Output of G_2 is 1

Let us assume $Q = 1$

Output of G_3 is 1

Output of G_4 is 0



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1
↑	1	0	1	1

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Case 6: Clk = 1; J = 1; K = 1;

Output of G_1 is Q

Output of G_2 is $\bar{Q}$

Output of G_3 is 1

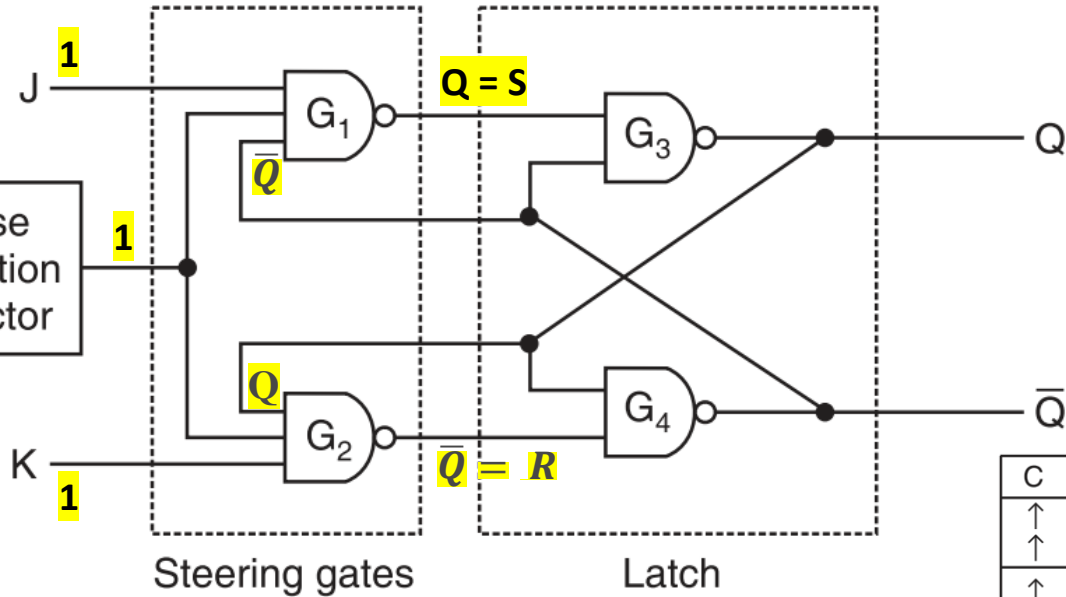
Output of G_4 is 0

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Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1
↑	1	0	1	1
↑	1	1	0	1
↑	1	1	1	0



JK Flip Flop



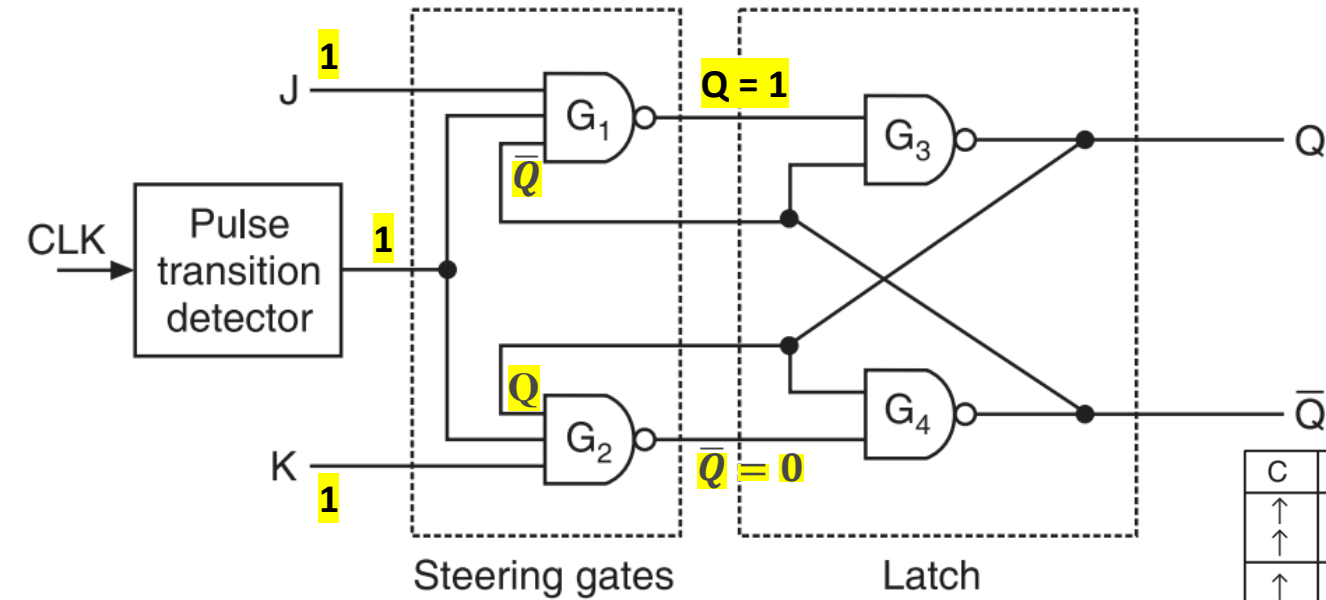
Case 7: Clk = 1; J = 1; K = 1;

Output of G_1 is Q

Output of G_2 is $\bar{Q}$

Output of G_3 is 0

Output of G_4 is 1



Truth Table of SR Latch

S	R	Q_{n+1}
0	0	Invalid State
0	1	1 (Set)
1	0	0 (Reset)
1	1	No Change (Hold State)

C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1
↑	1	0	1	1
↑	1	1	0	1
↑	1	1	1	0

JK Flip Flop

Truth Table of JK Flip Flop

CLK	J	K	Q_{n+1}
1 (Triggered)	0	0	No Change (Hold State)
1 (Triggered)	0	1	If Q =0; No change
1 (Triggered)	0	1	If Q =1; 0 (Reset)
1 (Triggered)	1	0	If Q =0 1 (Set)
1 (Triggered)	1	0	If Q = 1; No change
1 (Triggered)	1	1	Toggle

Characteristics Table of JK Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

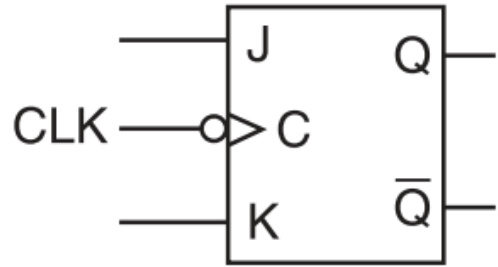
	KQ_n	00	01	11	10
J	0	0	1	0	0
	1	1	1	0	1

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

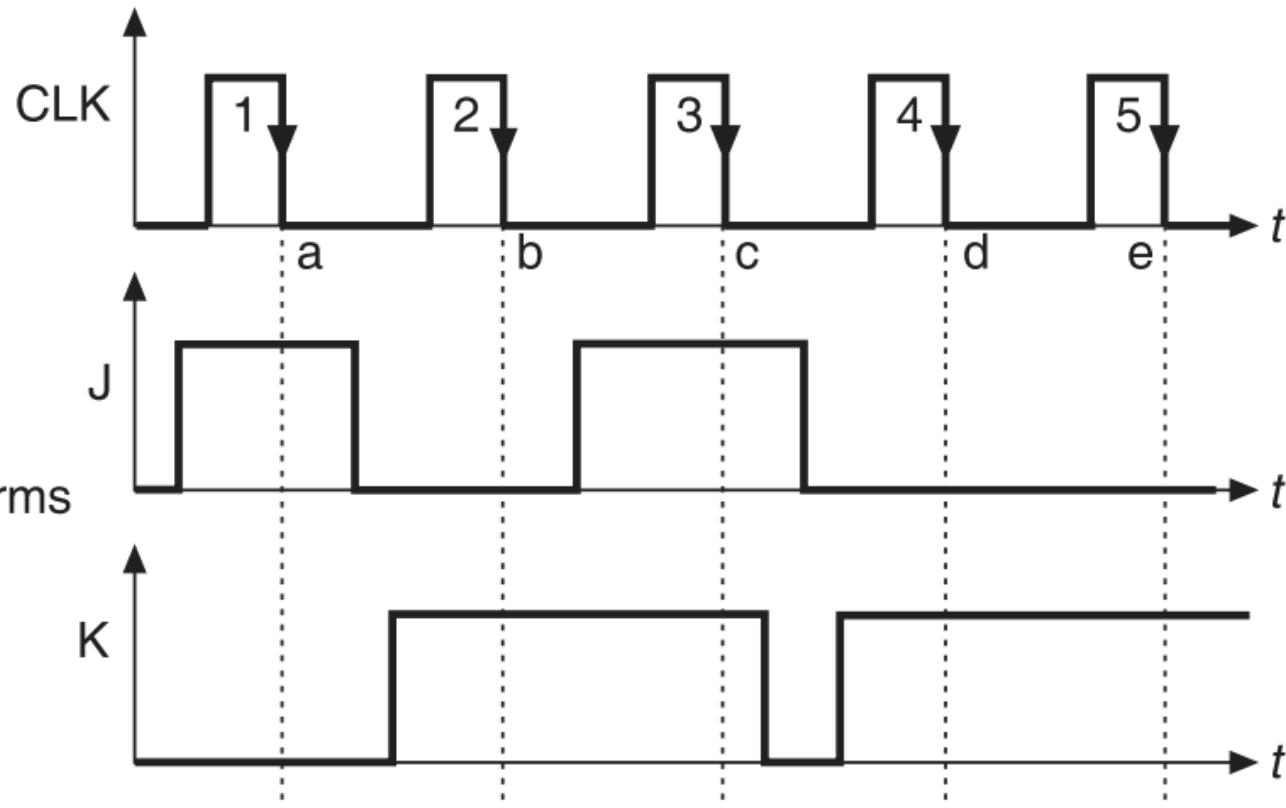
Negative Edge Triggered JK Flip Flop



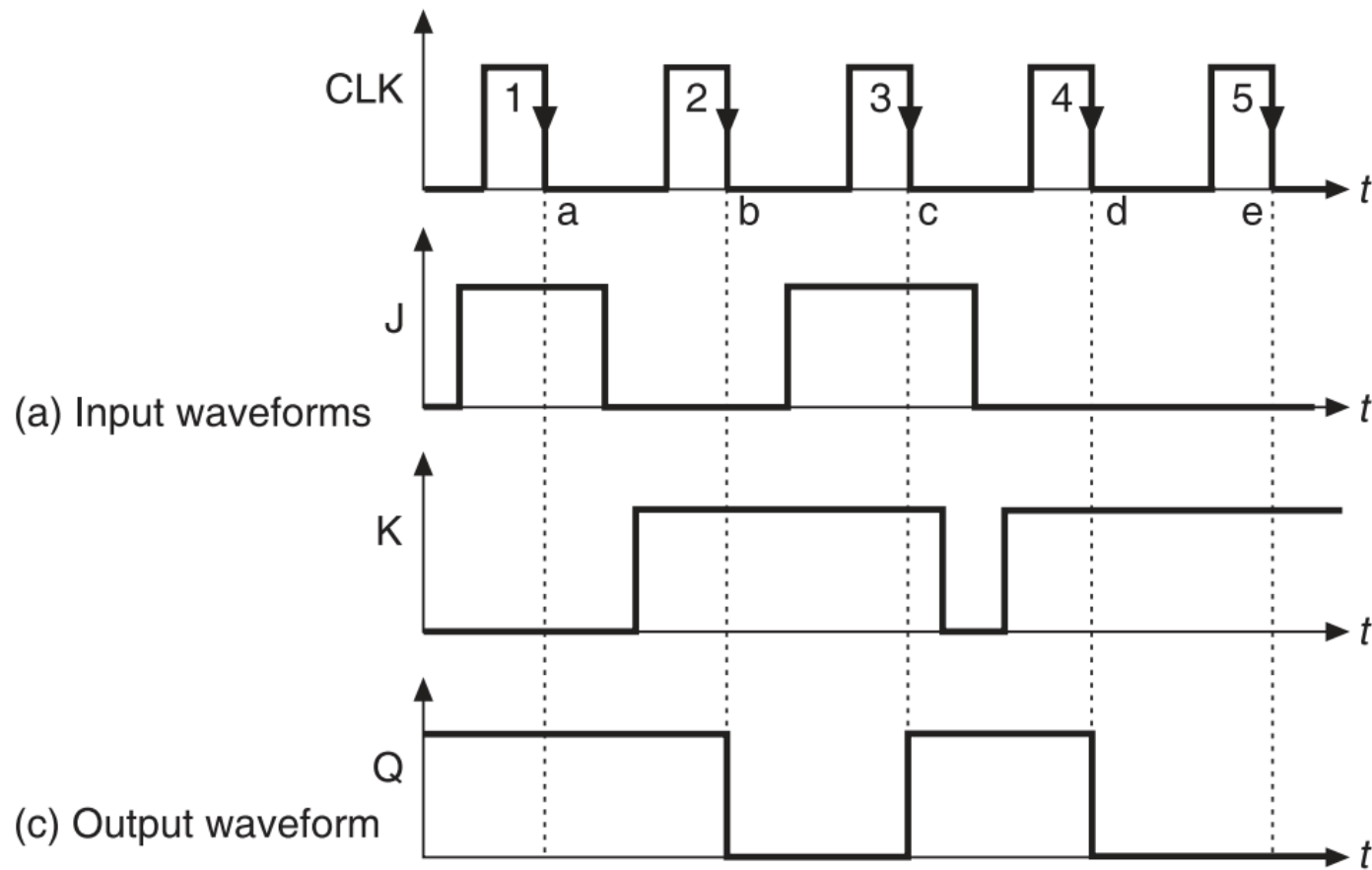
Problem: The waveforms shown in Figure are applied to the edge-triggered J-K flip-flop shown in Figure. Draw the output waveform.



(a) Input waveforms



Negative Edge Triggered JK Flip Flop



Truth Table of JK Flip Flop

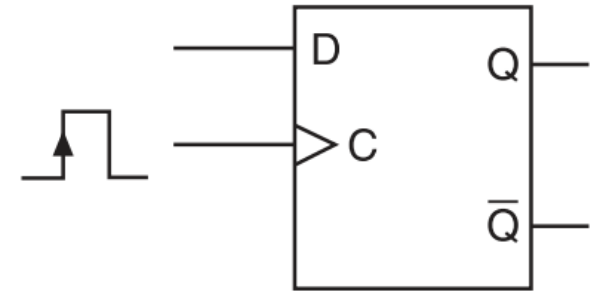
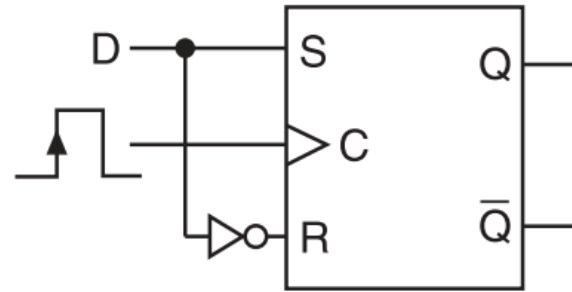
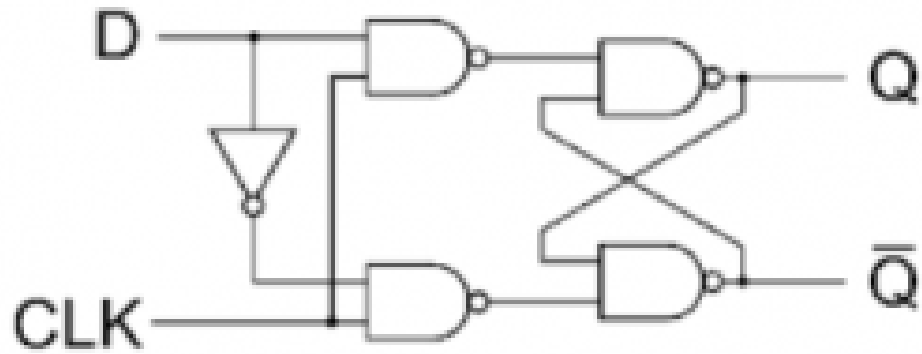
C	J	K	Q_n	Q_{n+1}
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1
↑	1	0	1	1
↑	1	1	0	1
↑	1	1	1	0
0	x	x	0	0
0	x	x	1	1

Negative Edge Triggered JK Flip Flop



1. Initially $J = 0$, $K = 0$ and $CLK = 0$. Assume that the initial state of the flip-flop is a 1, i.e. $Q = 1$ initially.
2. At the negative-going edge of the first clock pulse (i.e. at a), $J = 1$ and $K = 0$. So, Q remains as a 1 and, therefore, $\bar{Q}$ as a 0.
3. At the trailing edge of the second clock pulse (i.e. at b), $J = 0$ and $K = 1$. So, the flip-flop resets. That is, Q goes to a 0 and $\bar{Q}$ to a 1.
4. At the falling edge of the third clock pulse (i.e. at c), both J and K are a 1. So, the flip-flop toggles. That is, Q changes from a 0 to a 1 and $\bar{Q}$ from a 1 to a 0.
5. At the negative-going transition of the fourth clock pulse (i.e. at d), $J = 0$ and $K = 1$. So, the flip-flop RESETS, i.e. Q goes to a 0 and $\bar{Q}$ to a 1.
6. At the negative going edge of the fifth clock pulse (i.e. at d), $J = 0$ and $K = 1$. So the flip-flop remains reset, i.e. Q remains as 0 and $\bar{Q}$ remains as 1.

D Flip Flop



Truth Table of D Flip Flop

CLK	D	Q_{n+1}
0 (Not Triggered)	x	No Change (Hold State)
1 (Triggered)	0	0 (Reset)
1 (Triggered)	1	1 (Set)

Characteristics Table of D Flip Flop

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$$Q_{n+1} = D$$

Excitation Table of D Flip Flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1